MÖBIUS INVERSION THROUGH PARTIALLY ORDERED SETS

SAYAN KAR

CONTENTS

1. MÖBIUS FUNCTION

Definition 1.1. A *partially ordered set* (or *poset*) is a pair $(P, \preceq)$ where P is a set and $\preceq$ is a binary relation on P satisfying the following properties for all $x, y, z \in P$:

(1) **Reflexivity:** $x \preceq x$.
(2) **Antisymmetry:** if $x \preceq y$ and $y \preceq x$, then $x = y$.
(3) **Transitivity:** if $x \preceq y$ and $y \preceq z$, then $x \preceq z$.

Lemma 1.2. *Suppose that $|P| = n$. Then we can label the elements of $P = \{p_1, p_2, \dots, p_n\}$ so that*

$$p_i \preceq p_j \implies i \leq j.$$[1]

Proof. We prove this by induction on n. The base case $n = 1$ is trivial. Choose a maximal element of P and label it p_n. Let $P' = P \setminus \{p_n\}$. By the induction hypothesis, we can label $P' = \{p_1, p_2, \dots, p_{n-1}\}$ so that

$$p_i \preceq p_j \implies i \leq j.$$

Since p_n is maximal, this labeling together with p_n gives the required labeling of all of P. □

Let $(P, \preceq)$ be a finite partially ordered set. We define the function

$$\zeta : P^2 \to \{0, 1\}$$

by

$$\zeta(x, y) = \begin{cases} 1 & \text{if } x \preceq y, \\ 0 & \text{otherwise.} \end{cases}$$

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[1] This type of ordering of a poset is called a *topological ordering*.

Definition 1.3. Let P be a finite poset. Consider the matrix

$$A_\zeta = [\zeta(x,y)]_{x,y\in P}.$$

By Lemma 1.2, there is a topological ordering of P, the matrix A_ζ can an upper triangular matrix with all diagonal entries equal to 1. Hence A_ζ is invertible. Its inverse is denoted

$$A_\mu = [\mu(x,y)]_{x,y\in P}.$$

The function $\mu : P^2 \to \mathbb{Z}$ is called the *Möbius function* of the poset P.

Since $A_\mu A_\zeta = I$, we obtain

$$\sum_{z\in P} \mu(x,z)\zeta(z,y) = \sum_{x\preceq z\preceq y} \mu(x,z) = \delta(x,y).$$

Theorem 1.4. *Consider the two following cases:*

(a) *Let P be the set of subsets of a finite set X, ordered by inclusion $\subseteq$. Then*

$$\mu(A,B) = \begin{cases} (-1)^{|B\setminus A|} & \text{if } A\subseteq B, \\ 0 & \text{otherwise.} \end{cases}$$

(b) *Let $P = [n]$ with the order defined by $a \preceq b$ if a divides b. Then*

$$\mu(a,b) = \begin{cases} (-1)^r & \text{if } b/a \text{ is the product of } r \text{ distinct primes,} \\ 0 & \text{otherwise.} \end{cases}$$

Proof. We only need to verify the defining relation

$$\sum_{x\preceq z\preceq y} \mu(x,z) = \begin{cases} 1 & x = y, \\ 0 & x \neq y. \end{cases}$$

(a) If $A \subseteq B$, then

$$\sum_{A\subseteq C\subseteq B} (-1)^{|C\setminus A|} = \sum_{k=0}^{|B\setminus A|} \binom{|B\setminus A|}{k} (-1)^k = (1-1)^{|B\setminus A|}.$$

This equals 0 unless $A = B$, in which case it equals 1.

(b) Suppose

$$b/a = p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r},$$

where $p_1, p_2, \dots, p_r$ are distinct primes and $k_1, k_2, \dots, k_r \geq 1$. Then any c satisfying $a \mid c \mid b$ corresponds to choosing a subset of the primes $\{p_1, \dots, p_r\}$. Hence

$$\sum_{a|c|b} \mu(c,b) = \sum_{S\subseteq[r]} (-1)^{|S|} = \begin{cases} 1 & r = 0, \\ 0 & r \geq 1. \end{cases}$$

This verifies the required identity.

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2. Möbius Inversion

Theorem 2.1. *Let $(P, \preceq)$ be a finite poset and let $f, g, h : P \to \mathbb{R}$ be functions such that*

$$g(x) = \sum_{a \preceq x} f(a), \qquad h(x) = \sum_{b \succeq x} f(b). \tag{2.1}$$

Then the function f can be recovered from g or h via

$$f(x) = \sum_{a \preceq x} \mu(a, x) g(a), \qquad f(x) = \sum_{b \succeq x} \mu(x, b) h(b). \tag{2.2}$$

Proof. Since P is finite, we may label its elements as $P = \{p_1, \ldots, p_n\}$ in a way compatible with the partial order. Any function $f : P \to \mathbb{R}$ can then be identified with a column vector

$$\mathbf{f} = (f(p_1), f(p_2), \ldots, f(p_n))^T.$$

Similarly we associate vectors $\mathbf{g}$ and $\mathbf{h}$ to the functions g and h.

Let $A_\zeta = [\zeta(x, y)]$ be the matrix corresponding to the zeta function of the poset. The relation

$$g(x) = \sum_{a \preceq x} f(a)$$

means that each coordinate of $\mathbf{g}$ is obtained by summing the entries of $\mathbf{f}$ over elements below it in the order. In matrix form this is

$$\mathbf{g} = A_\zeta^T \mathbf{f}.$$

Likewise, the relation

$$h(x) = \sum_{b \succeq x} f(b)$$

corresponds to the matrix equation

$$\mathbf{h} = A_\zeta \mathbf{f}.$$

Since A_ζ is invertible and its inverse is A_μ, we obtain

$$\mathbf{f} = A_\zeta^{-T} \mathbf{g} = A_\mu^T \mathbf{g} \quad \text{and} \quad \mathbf{f} = A_\zeta^{-1} \mathbf{h} = A_\mu \mathbf{h}.$$

Writing these matrix identities co-ordinatewise yields the stated inversion formulas. $\square$

3. Applications of Möbius Inversion

3.1. **Inclusion–Exclusion Principle.** Möbius inversion gives a particularly elegant proof of the classical principle of inclusion–exclusion. Let $\{A_i\}_{i \in I}$ be a family of subsets of a finite set X. For $J \subseteq I$, define

$$f(J) = \left| \left(\bigcap_{i \in J} A_i \right) \cap \left(\bigcap_{i \notin J} (X \setminus A_i) \right) \right|,$$

that is, the number of elements that lie in exactly the sets indexed by J. Let

$$h(J) = \left| \bigcap_{i \in J} A_i \right|.$$

Then
$$h(J) = \sum_{K \supseteq J} f(K).$$

Applying Möbius inversion on the poset $(\mathcal{P}(I), \subseteq)$, we obtain
$$f(J) = \sum_{K \supseteq J} \mu(J,K) h(K) = \sum_{K \supseteq J} (-1)^{|K|-|J|} h(K).$$

Putting $J = \varnothing$ gives
$$\left| \bigcap_{i \in I} (X \setminus A_i) \right| = \sum_{K \subseteq I} (-1)^{|K|} \left| \bigcap_{j \in K} A_j \right|.$$

3.2. **Totient Function.** Möbius inversion gives another proof of the classical closed form for Euler's totient function. We briefly note that the divisibility relation on the natural numbers forms a partially ordered set. In this *divisibility poset*, we write $a \preceq b$ if a divides b. For a function $f : \mathbb{N} \to \mathbb{R}$, define
$$g(n) = \sum_{d|n} f(d).$$
Möbius inversion on this poset gives
$$f(n) = \sum_{d|n} \mu(d,n) g(d) = \sum_{\substack{d|n \\ n/d \text{ square free}}} (-1)^{p(n/d)}\, g(d),$$
where $p(m)$ denotes the number of distinct prime factors of m. For a natural number n, let $\varphi(n)$ denote the number of integers $m \leq n$ such that $\gcd(m,n) = 1$.

Lemma 3.1.
$$n = \sum_{d|n} \varphi(d) = \sum_{d|n} \varphi(n/d).$$

Proof. If $(m,n) = d$, then we can write $m = m_1 d$ and $n = n_1 d$ where $\gcd(m_1, n_1) = 1$. Hence the number of such m is the number of choices of m_1, which is $\varphi(n_1) = \varphi(n/d)$. $\square$

Applying Möbius inversion to this identity with $g(n) = n$ and $f(n) = \varphi(n)$ gives
$$\varphi(n) = \sum_{d|n} (-1)^{p(n/d)} d = \sum_{d|n} (-1)^{p(d)} \frac{n}{d}.$$
Factoring out n we obtain
$$\varphi(n) = n \sum_{d|n} \frac{(-1)^{p(d)}}{d}.$$

If $n = p_1^{k_1} p_2^{k_2} \cdots p_r^{k_r}$ where $p_1, p_2, \ldots, p_r$ are distinct primes and $k_1, k_2, \ldots, k_r \geq 1$, the multiplicative structure of the sum yields the familiar product form
$$\varphi(n) = n \prod_{i=1}^{r} \left(1 - \frac{1}{p_i}\right).$$

3.3. **Colored Necklace Problem.** Möbius inversion also appears naturally in many counting problems. A classical example is the problem of counting colored necklaces.

Suppose we have an infinite supply of beads of λ colours and wish to form a necklace with n beads. Any necklace can be represented by a sequence $(a_1, \ldots, a_n)$, where we identify sequences that differ by a cyclic permutation, since they represent the same necklace.

We say that a necklace of length n is *primitive* if it is not obtained by repeating a shorter necklace. More precisely, if $d \mid n$, a necklace is said to have period d if it is obtained by repeating n/d times a primitive necklace of length d.

First note that the number of sequences of length n is λ^n. Every such sequence arises from a primitive necklace of some period d dividing n. If $M(d)$ denotes the number of primitive necklaces of length d, then each such necklace gives rise to d distinct sequences depending on the choice of the starting position. Hence

$$\lambda^n = \sum_{d|n} d\, M(d).$$

Applying Möbius inversion to this identity gives

$$M(n) = \frac{1}{n} \sum_{d|n} \mu(d) \lambda^{n/d}.$$

The total number of necklaces of length n is therefore

$$N_n = \sum_{d|n} M(d).$$

Substituting the expression for $M(d)$, we obtain

$$N_n = \sum_{d|n} \sum_{\ell|d} \frac{1}{d} \mu(\ell, d) \lambda^{\ell}.$$

Writing $d = k\ell$ and rearranging the sums gives

$$N_n = \sum_{\ell|n} \frac{\lambda^\ell}{\ell} \sum_{k|n/\ell} \frac{\mu(1,k)}{k}.$$

Using the identity

$$\sum_{k|m} \frac{\mu(1,k)}{k} = \frac{\varphi(m)}{m},$$

we finally obtain the well-known formula

$$N_n = \frac{1}{n} \sum_{d|n} \varphi(n/d) \lambda^d.$$

Thus the number of necklaces of length n formed from λ colours is

$$N_n = \frac{1}{n} \sum_{d|n} \varphi(n/d) \lambda^d.$$

REFERENCES

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[2] S. M. Cioabă and M. Ram Murty, *A First Course in Graph Theory and Combinatorics*, Springer / Hindustan Book Agency, 2009.